Statistical Learning and Machine Learning

Exam Topics - May 2022

TOPIC 1

You are given a set of data points $\mathbf{x}_i \in \mathbb{R}^{10}$, $i = 1, \ldots, N$, each belonging to one of three classes $\mathcal{C}_k$, $k = 1, \ldots, 3$. You want to use a feedforward neural network formed by two hidden-layers for solving this classification problem.

Answer the following questions:

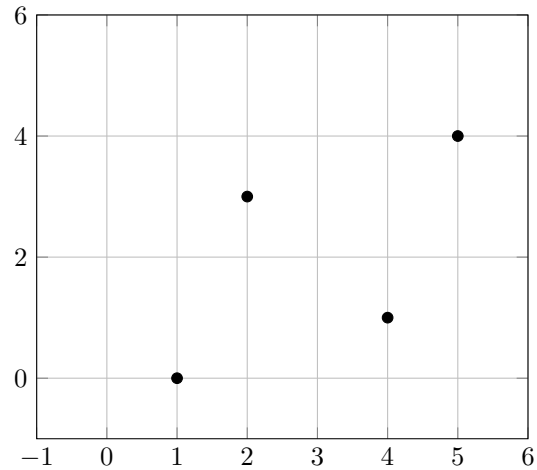
1) (Score: 20%) Describe which are the parameters and the hyper-parameters of the neural network. Use appropriate mathematical symbols (e.g., matrices, functions).
2) (Score: 20%) Draw a neural network that can be used to solve the above-described classification problem. Make appropriate decisions for all the hyper-parameters of the neural network. In your drawing include also the parameters of the neural network and the activation functions.
3) (Score: 20%) Describe how to estimate the values of the neural network parameters (including the optimization function), and its hyper-parameters.
4) (Score: 20%) Write down which optimization problem you will use to optimize the network's parameters, and describe all terms of the problem (including the dimensions of each matrix/vector).
5) (Score: 20%) Show that the use of the linear activation function (for all layers) makes the above network equivalent to a two-layer (no hidden layers) network. Based on the above, describe why it is important to use non-linear activation functions in neural networks.

TOPIC 2

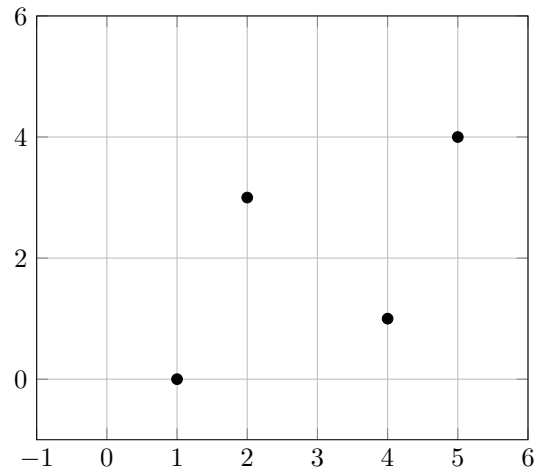
Consider a data set formed by four data points $\mathbf{x}_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ and $\mathbf{x}_4 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

We want to represent the data in only one dimension, so we turn to principal components analysis (PCA).

1) (Score: 30%) Compute the two (unit-length) principal components of the matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4]$, and state which one the Principal Component Analysis (PCA) algorithm would choose if you request just one principal component. Please provide an exact answer, without approximation.
2) (Score: 20%) The plot below depicts the sample points from $\mathbf{X}$. We want an one-dimensional representation of the data obtained by applying PCA, so draw the principal component direction (as a line) and the projections of all four data points onto that principal component direction. Label each projected point with its principal coordinate value (where the origin's principal coordinate is zero). Give the principal coordinate values exactly.



3) (Score: 30%) Consider that the data points belong to two classes: $\mathbf{x}_1$ and $\mathbf{x}_3$ belong to class one, and $\mathbf{x}_2$ and $\mathbf{x}_4$ belong to class two. Compute the discriminant axis (direction) which is the result of applying Linear Discriminant Analysis (LDA). Please provide an exact answer, without approximation.
4) (Score: 20%) The plot below depicts the sample points from $\mathbf{X}$. We want an one-dimensional representation of the data obtained by applying LDA, so draw the discriminant axis (direction, as a line) and the projections of all four data points onto that axis.



TOPIC 3

The two classes of a binary classification problem are formed by the blue and red samples plotted in Figure 1.

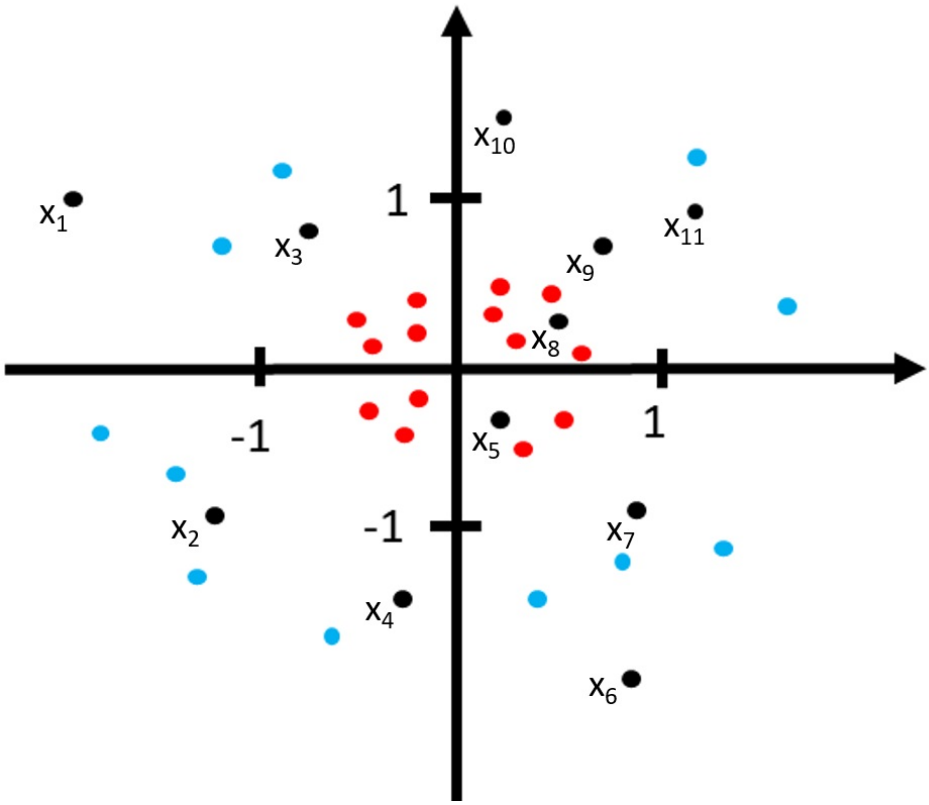


Fig. 1. 2-dimensional data points plotted as dots. Data belonging to class 1 are illustrated in blue color. Data belonging to class 2 are illustrated in red color. Test data $\mathbf{x}_i$, $i = 1, \ldots, 11$ are illustrated in black color.

Answer the following questions:

1) (Score: 20%) Would you use Linear Regression (using an indicator matrix) to classify data from the two classes? If yes, why? If no, why not?
2) (Score: 20%) Can you use a Support Vector Machine to classify data from the two classes? If yes, how this can be done? If no, why not?
3) (Score: 20%) There exists a data transformation $y_i = \phi(\mathbf{x}_i)$ such that the Support Vector Machine (with linear kernel function) applied on the transformed data can provide 100% training classification performance. Write down the data transformation function $y_i = \phi(\mathbf{x}_i)$.
4) (Score: 20%) Draw the transformed data and the decision function of the Support Vector Machine (linear kernel function) in the new feature space and the corresponding decision function in the original feature space. When drawing the transformed data, no exact values of their dimension(s) are needed, just the rough arrangement of them in the new feature space. (You can capture the figure in a screenshot and use a drawing program, like Paint)
5) (Score: 20%) Which are the Support Vectors (a) in the original feature space, (b) in the transformed feature space?

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TOPIC 4

Consider a training data set of inputs $\{t_1 = -1, t_2 = 1, t_3 = 2\}$ and the corresponding targets or outputs $\{x_1 = 1, x_2 = 1.5, x_3 = 3\}$. The goal is to first fit an appropriate linear regression model to the above data using the least squares approach and then use that model to predict the value of x for a new value of t.

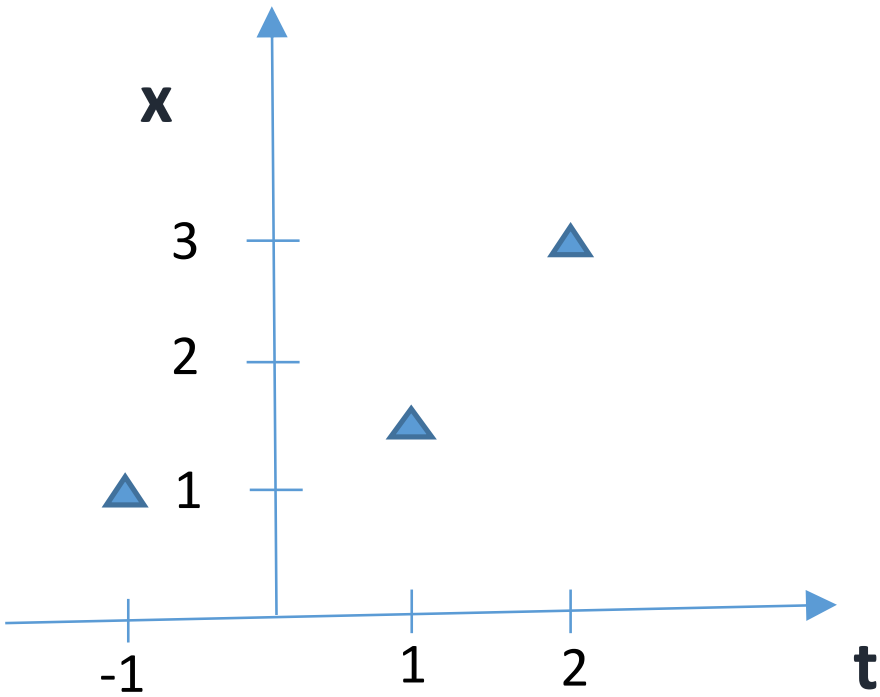

Fig. 2. The training data set consisting of $N = 3$ points (represented as blue triangles) plotted in the 2D space.

1) (Score: 25 %) Considering the following choice for the basis functions $\mathbf{\Phi}(t) = \{\Phi_0(t), \Phi_1(t)\} = \{1, t\}$ that leads to the simplest linear model for regression $x(t, w) = w_0 + w_1 t$, write down the corresponding *normal equations* for the least squares problem.
2) (Score: 25 %) Solve the normal equations to obtain the parameters or the weights (w_0 and w_1) of the linear model. Hint: The inverse of a 2×2 matrix is
$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$
3) (Score: 25 %) Plot the linear model function $x(t, w) = w_0 + w_1 t$, obtained in the previous step, in Fig. 2. You can assess your model by its ability to fit the training data i.e., how close it is to the training data.
4) (Score: 25 %) Using the linear model that is estimated in the previous steps, predict the value of x for $t = 0$, and for $t = 3$. Plot the corresponding points in Fig. 2 as crosses.